3AL Hand-in 2 2024 Solutions

Due: Monday 8 April 11:00 pm

Please make sure all of your answers are clearly labelled.

1. Determine whether each statement is true or false. No explanation needed.

   (a) Let X be a G-set and $x \in X$. Then $x \in X_f$ if and only if $|G : S(x)| = 1$.

   (b) If G is a group acting on a set X, then $|G| > |X|$.

   (c) If X is a G-set with $x \in X$, then $G * x$ is a subgroup of $S(x)$.

   (d) Let G be a group. If the group action of conjugation (by G on G) is the trivial group action, then G is abelian.

(a) True. We have $x \in X_f$ if and only if $G * x = \{x\}$. But $|G * x| = |G : S(x)|$. So, $G * x = \{x\}$ if and only if $|G : S(x)| = 1$.

(b) False. Note that it isn't necessarily true that $|G| < |X|$ either. A group of any size can act on a set of any size (e.g. we can always just take the trivial action).

(c) False. The orbit $G * x$ is a subset of X, it doesn't necessarily have any group structure.

(d) True. If $gxg^{-1} = x$ for all $x, g \in G$, then we have $gx = xg$ for all $x, g \in G$. That is, G is abelian.

[4 marks]

2. Let G be a group and $\mathrm{Aut}(G)$ be the group of automorphisms of G.

   (a) Prove that $* : \mathrm{Aut}(G) \times G \to G$ defined by $\varphi * g = \varphi(g)$ is a group action.

   (b) Find $S(1_G)$.

(a) The identity in $\mathrm{Aut}(G)$ is id_G. We have $\mathrm{id}_G * g = \mathrm{id}_G(g) = g$ for all $g \in G$. For $\varphi, \psi \in \mathrm{Aut}(G)$, we have

$$\varphi * (\psi * g) = \varphi * \psi(g) = \varphi(\psi(g)) = (\varphi \circ \psi) * g.$$

Thus $*$ is a group action.

Note: Please subtract a mark if student writes $\varphi * \psi * g$.

(b) $S(1_G) = \mathrm{Aut}(G)$.

[3 + 1 = 4 marks]

3. Let G be a finite group acting on a set X and let $x \in X$. Prove or disprove: The number of elements in $G * x$ divides $|G|$.

The statement is true. By the Orbit-Stabiliser Theorem, $|G * x| = |G : S(x)|$. Since G is a finite group, we have that $|G : S(x)| = |G|/|S(x)|$ must divide $|G|$. Thus $|G * x| \mid |G|$.

[4 marks]

Maximum mark = 10

Available marks = 12